Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **Java/Python services**, **Spring Boot APIs**, secure data pipelines, automated tests, and **resilient financial workflows** across production systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Achieved 99%+ reliability across 5K+ monthly jobs through **data validation and reconciliation**, error handling, cloud storage, CI/CD releases, and production monitoring supporting dependable operational workflows and internal users reliably.
- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant analytics platform supporting 500+ users.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during concurrent processing and sustained production traffic reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Increased pipeline efficiency by 68% through **AWS Glue and S3 ETL**, transforming JSON, CSV, and Parquet inputs into normalized datasets while reducing storage costs by 15% reliably.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst - Research Fellowship | *India*
- Achieved 96%+ classification accuracy across 650K+ URLs through a **reproducible detection pipeline**, comparing Random Forest, CNN, RNN, LSTM, Bi-LSTM, and GRU models for phishing and malware identification.
- Engineered 21 behavioral indicators covering URL length, IP presence, and suspicious keywords, translating noisy inputs into **validated risk features** that exposed phishing and malware patterns reliably at scale.

LST Capitals | *Web Developer Intern* | *India* **November 2022 – January 2023**
- Delivered **sub-200ms Flask APIs** across 50K+ booking records, implementing pricing, validation, payment webhooks, Firebase synchronization, and database transactions while preventing conflicting reservations across concurrent customer workflows reliably.
- Reduced manual refund handling by approximately 70% through **payment, refund, and dispute workflows**, integrating webhook processing, real-time status synchronization, authentication, and transactional controls across customer booking operations.

Skills

Languages: Java, Python, SQL, TypeScript, JavaScript, C++, Bash
Backend & APIs: Spring Boot, FastAPI, Flask, Node.js, REST APIs, Microservices, JSON, Webhooks, JWT, Object-Oriented Programming
Data & Financial Systems: PostgreSQL, MySQL, MongoDB, AWS RDS, Firebase, ETL, Data Pipelines, Data Modeling, Schema Design, Query Optimization, Transaction Processing, Payment Workflows
Security & Risk: Phishing Detection, Malware Detection, Classification, Feature Engineering, Role-Based Access Control, Token-Based Access Controls, Authentication, Data Validation, Risk Analysis
Cloud & Delivery: AWS, Azure, AWS Glue, Amazon S3, Docker, Kubernetes, Jenkins, Git, GitHub, CI/CD
Quality & SDLC: Software Development Lifecycle, Unit Testing, Integration Testing, Automated Testing, PyTest, Mockito, Debugging, Monitoring, Error Handling, Fault Isolation, Technical Documentation, Agile, Scrum

Projects

Financial Crimes Case Management - Design Concept
- Designed a **financial-crimes case platform** concept ingesting transaction alerts into explainable risk rules, analyst queues, evidence timelines, role-based controls, audit logs, disposition workflows, and regulator-ready case reports.

FinNexus | Live
- Built a **secure fintech platform** supporting 50K+ users and 1M+ transactions with real-time synchronization, advanced filtering, token-based access controls, MongoDB-backed data retrieval, Plaid integration, and Kubernetes deployment.

Chat2SQL | Live
- Extended a **Java and TypeScript database workspace** supporting 10+ database systems, using Docker, Maven, Node.js, and Yarn for cross-platform schema exploration, query workflows, local debugging, and deployment.